PADHAKU
Mathematics · Class 10 Mathematics

CHAPTER 2

Polynomials

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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❓ In-text question · Example 1 (Why?)

Why? **In Example 1 (Fig. 2.9), why is the number of zeroes 3 in (iii), 1 in (iv), 1 in (v) and 4 in (vi)?**

✔ ANSWER

The zeroes of p(x) are the x-coordinates of the points where the graph of y = p(x) meets the x-axis.

So the number of zeroes = number of points where the graph cuts or touches the x-axis.

- (iii) The graph crosses the x-axis at 3 points → **3 zeroes**.
- (iv) The graph meets the x-axis at only 1 point → **1 zero**.
- (v) The graph meets the x-axis at only 1 point → **1 zero**.
- (vi) The graph meets the x-axis at 4 points → **4 zeroes**.

📝 Exercise 2.1

1 **The graphs of y = p(x) are given in Fig. 2.10, for some polynomials p(x). Find the number of zeroes of p(x) in each case.**

✔ ANSWER

The number of zeroes of p(x) = the number of points where the graph of y = p(x) intersects (cuts or touches) the x-axis.

- (i) The graph is a line parallel to the x-axis; it does not meet the x-axis → **no zero (0 zeroes)**.
- (ii) The graph meets the x-axis at 1 point → **1 zero**.
- (iii) The graph meets the x-axis at 3 points → **3 zeroes**.
- (iv) The graph meets the x-axis at 2 points → **2 zeroes**.
- (v) The graph meets the x-axis at 4 points → **4 zeroes**.
- (vi) The graph meets the x-axis at 3 points → **3 zeroes**.

📝 Exercise 2.2

1 (i) **Find the zeroes of $x^2 - 2x - 8$ and verify the relationship between the zeroes and the coefficients.**

✔ ANSWER

$x^2 - 2x - 8 = x^2 - 4x + 2x - 8 = x(x - 4) + 2(x - 4) = (x - 4)(x + 2)$

$p(x) = 0$ when $x - 4 = 0$ or $x + 2 = 0$, i.e. $x = 4$ or $x = -2$.

Zeroes: 4 and −2

Here $a = 1$, $b = -2$, $c = -8$.

Sum of zeroes $= 4 + (-2) = 2 = -(-2)/1 = -b/a$ ✓

Product of zeroes = 4 × (−2) = −8 = −8/1 = c/a ✓

1 (ii) **Find the zeroes of $4s^2 - 4s + 1$ and verify the relationship between the zeroes and the coefficients.**

✔ ANSWER

$4s^2 - 4s + 1 = 4s^2 - 2s - 2s + 1 = 2s(2s - 1) - 1(2s - 1) = (2s - 1)(2s - 1)$

$p(s) = 0$ when $2s - 1 = 0$, i.e. $s = ½$ (twice).

Zeroes: ½ and ½

Here $a = 4$, $b = -4$, $c = 1$.

Sum of zeroes = ½ + ½ = 1 = −(−4)/4 = −b/a ✓

Product of zeroes = ½ × ½ = ¼ = 1/4 = c/a ✓

1 (iii) **Find the zeroes of $6x^2 - 3 - 7x$ and verify the relationship between the zeroes and the coefficients.**

✔ ANSWER

Write in standard form: $6x^2 - 7x - 3$.

$6x^2 - 7x - 3 = 6x^2 - 9x + 2x - 3 = 3x(2x - 3) + 1(2x - 3) = (3x + 1)(2x - 3)$

$p(x) = 0$ when $3x + 1 = 0$ or $2x - 3 = 0$, i.e. $x = -1/3$ or $x = 3/2$.

Zeroes: −1/3 and 3/2

Here $a = 6$, $b = -7$, $c = -3$.

Sum of zeroes = −1/3 + 3/2 = (−2 + 9)/6 = 7/6 = −(−7)/6 = −b/a ✓

Product of zeroes = $(-1/3) \times (3/2) = -1/2 = -3/6 = c/a$ ✓

1 (iv) **Find the zeroes of $4u^2 + 8u$ and verify the relationship between the zeroes and the coefficients.**

✔ ANSWER

$4u^2 + 8u = 4u(u + 2)$

$p(u) = 0$ when $4u = 0$ or $u + 2 = 0$, i.e. $u = 0$ or $u = -2$.

Zeroes: 0 and −2

Here $a = 4$, $b = 8$, $c = 0$.

Sum of zeroes $= 0 + (-2) = -2 = -8/4 = -b/a$ ✓

Product of zeroes $= 0 \times (-2) = 0 = 0/4 = c/a$ ✓

1 (v) **Find the zeroes of $t^2 - 15$ and verify the relationship between the zeroes and the coefficients.**

✔ ANSWER

$t^2 - 15 = t^2 - (\sqrt{15})^2 = (t - \sqrt{15})(t + \sqrt{15})$

$p(t) = 0$ when $t = \sqrt{15}$ or $t = -\sqrt{15}$.

Zeroes: $\sqrt{15}$ and $-\sqrt{15}$

Here $a = 1$, $b = 0$, $c = -15$.

Sum of zeroes $= \sqrt{15} + (-\sqrt{15}) = 0 = -0/1 = -b/a$ ✓

Product of zeroes $= \sqrt{15} \times (-\sqrt{15}) = -15 = -15/1 = c/a$ ✓

1 (vi) **Find the zeroes of $3x^2 - x - 4$ and verify the relationship between the zeroes and the coefficients.**

✔ ANSWER

$3x^2 - x - 4 = 3x^2 - 4x + 3x - 4 = x(3x - 4) + 1(3x - 4) = (3x - 4)(x + 1)$

$p(x) = 0$ when $3x - 4 = 0$ or $x + 1 = 0$, i.e. $x = 4/3$ or $x = -1$.

Zeroes: 4/3 and −1

Here $a = 3$, $b = -1$, $c = -4$.

Sum of zeroes $= 4/3 + (-1) = 1/3 = -(-1)/3 = -b/a$ ✓

Product of zeroes $= (4/3) \times (-1) = -4/3 = c/a$ ✓

2 (i) **Find a quadratic polynomial whose sum and product of zeroes are 1/4 and −1 respectively.**

✔ ANSWER

A quadratic polynomial with zeroes α, β is $k[x^2 - (\alpha + \beta)x + \alpha\beta]$.

$\alpha + \beta = 1/4$, $\alpha\beta = -1$

Polynomial $= x^2 - (1/4)x - 1$

Multiply by 4 ($k = 4$) to remove the fraction:

$4x^2 - x - 4$

2 (ii) **Find a quadratic polynomial whose sum and product of zeroes are √2 and 1/3 respectively.**

✔ ANSWER

$\alpha + \beta = \sqrt{2}$, $\alpha\beta = 1/3$

Polynomial = $x^2 - \sqrt{2}x + 1/3$

Multiply by 3 (k = 3):

$3x^2 - 3\sqrt{2}x + 1$

2 (iii) **Find a quadratic polynomial whose sum and product of zeroes are 0 and √5 respectively.**

✔ ANSWER

$\alpha + \beta = 0$, $\alpha\beta = \sqrt{5}$

Polynomial = $x^2 - 0 \cdot x + \sqrt{5}$

$x^2 + \sqrt{5}$

2 (iv) **Find a quadratic polynomial whose sum and product of zeroes are 1 and 1 respectively.**

✔ ANSWER

$\alpha + \beta = 1$, $\alpha\beta = 1$

Polynomial = $x^2 - 1 \cdot x + 1$

$x^2 - x + 1$

2 (v) **Find a quadratic polynomial whose sum and product of zeroes are −1/4 and 1/4 respectively.**

✔ ANSWER

$\alpha + \beta = -1/4$, $\alpha\beta = 1/4$

Polynomial = $x^2 - (-1/4)x + 1/4 = x^2 + (1/4)x + 1/4$

Multiply by 4 (k = 4):

$4x^2 + x + 1$

2 (vi) **Find a quadratic polynomial whose sum and product of zeroes are 4 and 1 respectively.**

✔ ANSWER

$\alpha + \beta = 4$, $\alpha\beta = 1$

Polynomial = $x^2 - 4x + 1$

$x^2 - 4x + 1$

🎉 **Chapter 2 done! Revise the answers once more tomorrow.**